Abdennour Kerboua

AI Researcher • Computer Vision, NLP & Multimodal Learning

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Professional Experience

IDEMIA Public Security **Apr 2026-Present**
Research Engineer, Deep Learning Intern *Courbevoie, France*

- Surveyed state-of-the-art methods for object, person, and luggage re-identification (Re-ID), identifying methodology gaps and defining experimental priorities.
- Designed and implemented a robust luggage Re-ID baseline and benchmarking framework, enabling reproducible comparisons and systematic iteration.
- Developed Transformer-based, multi-view-aware and 3D-informed feature extraction and processing modules that achieved state-of-the-art results and materially improved accuracy over the baseline.
- Prepared research components for direct integration into IDEMIA products, including scalable data-handling workflows for experiments spanning approximately 20 million images, reproducibility, code and model versioning, and maintainable delivery.

ANFR - French National Frequency Agency **Aug-Sep 2024**
Radio Frequency Technician, Paris 2024 Olympic and Paralympic Games *Paris, France*

- Monitored and regulated the radio-frequency spectrum to support interference-free operations during the Paris 2024 Olympic and Paralympic Games.
- Audited, tested, and certified critical radio-frequency equipment for compliance with strict operational standards.

Education

École Normale Supérieure Paris-Saclay **Sep 2025-Dec 2026**
MSc, Mathematics, Vision, Learning (MVA) - GPA: 4.0/4.0 *Paris-Saclay, France*
Highly selective graduate program in AI research covering computer vision, machine learning, generative modeling, graph learning, NLP, optimization, robotics, and medical imaging.

Télécom Paris **Sep 2023-Dec 2026**
Engineering Degree (BSc + MSc equivalent), Computer Science and AI - GPA: 4.0/4.0 *Palaiseau, France*
Coursework in machine learning, deep learning, NLP, signal and image processing, 3D vision, statistics, optimization, and databases.

Lycée Janson-de-Sailly **Sep 2020-Jun 2023**
Preparatory Classes (CPGE), MPSI/MP - GPA: 4.0/4.0* *Paris, France*
Intensive curriculum in mathematics, physics, and computer science for competitive engineering-school admissions.

Selected Projects

referAI - AI Referee Perception System GitHub

- Fine-tuned YOLO11 on football data for per-frame person detection and applied multi-object tracking to maintain persistent trajectories.
- Trained a SoccerNet classifier for players, goalkeepers, and referees, using class-imbalance-aware validation and trajectory-level temporal aggregation for stable predictions.
- Reproducibly evaluated detection, tracking, and classification with structured exports and annotated videos. Next: calibration, team and ball tracking, possession, offside, and foul detection.

Cloud Shaper - Reinforcement Learning GitHub

- Built a Gymnasium environment for 2D cloud shaping under dynamic weather, with multi-channel spatial maps and local condensation/evaporation actions.
- Trained a CNN-based PPO actor-critic with procedural curriculum pretraining; achieved 0.731 on precision/recall, overflow, and contour alignment, optimized for CPU and Intel Arc XPU.

arXiv Evidence Agent GitHub

- Built a local, evidence-grounded RAG assistant to query and compare arXiv papers, using Docling for PDF extraction, multilingual E5 embeddings, and Qdrant vector search.
- Orchestrated a LangGraph agent pipeline with per-paper planning, hybrid dense and BM25 retrieval, reciprocal-rank fusion, reranking, and local Qwen 2.5 7B inference through OpenVINO.
- Containerized FastAPI, Qdrant, and model services with Docker Compose; delivered a source-citing UI and reproducible 30-case benchmark.
- Next: VLM-based scientific-image analysis, SFT, and preference alignment toward a reliable assistant that explains, compares, and cites scientific text and image publications.

Molecular Graph Captioning - ALTeGraD Challenge GitHub

- Designed a multimodal deep-learning framework connecting geometric deep learning and natural-language processing for automated molecular-graph captioning.
- Built and trained a dual-encoder architecture combining a Graph Convolutional Network (GCN) for molecular structures with SciBERT for scientific text.
- Applied contrastive representation alignment to map graph and text embeddings into a shared semantic space for caption generation.

NCSN and Diffusion Models for Medical Imaging GitHub

- Implemented training and evaluation workflows for Noise Conditional Score Networks (NCSN) and Denoising Diffusion Probabilistic Models (DDPM) on the MedMNIST benchmark.
- Evaluated both image generation and inpainting capabilities, comparing model behavior across medical-imaging conditions and data constraints.
- Assessed trade-offs between score-based and diffusion methods for medical-image synthesis.

Real-Time Sign Language Recognition GitHub

- Developed a real-time sign-language recognition prototype that processes live visual input and extracts hand landmarks with MediaPipe.
- Converted landmark coordinates into features and trained a Random Forest classifier for low-latency gesture prediction.

IBM x Télécom AI Challenge - Fraud Detection GitHub

- Built a fraud-detection pipeline for 300,000 records with data cleaning, feature engineering, and Random Forests, addressing severe class imbalance.
- Achieved a 69% F1-score and ranked 3rd of 25 teams in the IBM x Télécom AI Challenge.

Relevant Coursework

MVA - ENS Paris-Saclay **2025-2026**

- **Text, graphs, and multimodality:** NLP/LLMs, RAG, speech/audio processing (ASR, speech-to-text), graph learning, and multimodal learning.
- **Generative and representation learning:** graphical models, diffusion, score matching, VAEs/HVAEs, GANs/WGANs, self-supervised learning, SimCLR, Barlow Twins, CLIP, and distillation.
- **Computer vision and medical imaging:** ViT, YOLO, transfer learning, domain adaptation, multitask learning, Transformers, variational autoencoders, disentanglement, interpretability, and explainability.

Télécom Paris **2024-2025**

- **Machine learning and AI:** supervised and unsupervised learning, SVMs, Random Forests, clustering, LDA/QDA, Naive Bayes, KNN, neural networks, deep learning, text mining, NLP, speech and audio processing (ASR, speech-to-text), and generative models.
- **Image and video processing:** segmentation, restoration, denoising, registration, optimization, computational photography, 3D vision, motion estimation, tracking, calibration, remote sensing, and industrial applications.

Technical Skills

Research: Computer vision, NLP/LLMs, multimodal learning, graph machine learning, generative modeling, object re-identification, 3D/multi-view learning, representation learning, and medical imaging.
Methods: Transformers, SciBERT, GCNs, contrastive learning, RAG, LangGraph, RLHF, DPO, GRPO, diffusion models (DDPM, NCSN), ViT, YOLO, GANs, VAEs, and Random Forests.
Engineering: Python, Docker, Qdrant, Git, reproducible ML, 20M-image data, and deployment.

Languages: **French:** Native **English:** C1, full professional proficiency